Alejandro Granados

CTO at Heligráfics · Software architect · Valencia

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Data and AI architecture for wildfire fighting, forest management, emergencies and asset monitoring.

Profile

I started developing software in 2015, and since 2021 I have worked on AI applied to aerial imagery, geospatial data and emergencies, taking models from research prototypes to systems that work in the field.

As CTO I define the architecture, the engineering standards and regulatory compliance, and I train teams to design software, not just to write it.

Experience: 10+ years · Master's in AI (UPV)
Leadership: 7 teams · ~28 people · POs and Tech Leads
Specialty: Data management and AI in production
Methodology: DDD · Hexagonal architecture · SOLID
Core stack: Python · PostgreSQL / PostGIS · GCP · Kubernetes · Claude
Milestones: ISO 42001 · AI on terabytes of aerial imagery · From AI Scientist to CTO

Experience

CTO · Heligráfics, Valencia — Jan 2026 – Present

I lead the company's technology: strategy, architecture, teams and regulatory compliance (ISO 27001 and ENS).

- Defined the company's technology strategy and its development and architecture standards.
- Designed the domain data products of each business line, and I coordinate their development and deployment.
- I coordinate 7 teams of about 4 people, with their Product Owners and Tech Leads, and how they work: tasks, projects and ceremonies. I take part in hiring technical profiles and work with the management board.
- Chose the AI-assisted development tool through an A/B test between Claude and GitHub Copilot.
- Reduced infrastructure costs on GCP.

Staff AI Engineer · Heligráfics, Valencia — Feb 2024 – Dec 2025

Took the AI products for forest management to production and managed five simultaneous projects with cross-functional teams.

- Put into production quality analysis systems for forest plantations that process terabytes of aerial imagery.
- Led the ISO 42001 certification by implementing the data and AI management procedures and standards: one of the first companies in Spain to obtain it.
- Built internal libraries for data management and infrastructure as code.
- Migrated the company from GitLab to GitHub and rolled out CI/CD across every team.

Senior AI Engineer · Heligráfics, Valencia — Sep 2023 – Feb 2024

Turned AI prototypes into products in production. Promoted to Staff AI Engineer within five months.

- Deployed an application for two-way data transfer between aircraft and control centers, built for intermittent connectivity.
- Designed an internal MLOps platform to manage datasets, model versions and experiments.
- Took the aerial imagery detection and image segmentation prototypes for wildfire fighting to production.

Data Engineer · New Work, Valencia — Sep 2022 – Sep 2023

Distributed event-driven data pipelines, in a team with Scala as its main language.

- Built Kafka pipelines for user events with GDPR-compliant cascade deletion.
- Developed a sentiment classifier with language models.
- Was the internal Python reference for the AI and LLM initiatives.

AI Scientist · Heligráfics, Valencia — Jul 2021 – Sep 2022

Computer vision on aerial imagery, implementing architectures from research papers.

- Deployed a model that predicts burned areas from aerial imagery.
- Improved the internal libraries for parallel execution.
- Started the defect detection project for electrical towers.

Early career · Valencia 2015 – 2021

Full-Stack Developer · Dekalabs. Web applications and image generation prototypes (VAE and GAN). Mar 2020 – Jul 2021

Software Engineer · Praxya. ERP integrations and data migrations with Python and Odoo. Oct 2019 – Mar 2020

Junior Software Engineer · Visiion-Solutec. ERP solutions and distribution algorithms, with full ownership of the projects. Aug 2017 – Jun 2019

Independent developer · Own clients. Custom websites and applications, from requirements to delivery. 2015 – Aug 2017

Education

Master's · Artificial Intelligence, Pattern Recognition and Digital Imaging · Universitat Politècnica de València

Bachelor's · Computer Engineering · Universitat Politècnica de València
Final project: machine learning applied to genetic mutations.

Honors: Machine Learning, Data Structures and Algorithms, Software Engineering, Databases, Mathematical Analysis

Skills

Data management: Data Mesh (Data as a Product), Data pipelines (Airflow, Apache Spark), Event streaming (Kafka), Relational databases (PostgreSQL), Columnar storage (Parquet)
AI and MLOps: Computer vision & image segmentation (U-Net, ViT), Deep learning (PyTorch), MLOps (Vertex AI, Kubeflow), Model serving (ONNX)
Language models: LLMs (Claude), RAG, AI agents (MCP), LLM evaluation
Geospatial: Spatial databases (PostGIS, SpatiaLite), Raster processing (GDAL, Cloud Optimized GeoTIFF), Map services (GeoServer), Python geo (GeoPandas, Rasterio)
Real time and video: IoT messaging (MQTT, EMQX), Real-time web (WebSockets), Video streaming (WebRTC, HLS, RTMP, Oven Media Engine)
Backend: APIs (FastAPI, Django), Database migrations (Alembic), Node.js, Frontend (React), Microservices
Infrastructure: GCP (Cloud Run, Cloud Storage), Containers (Docker, Kubernetes), Infrastructure as code (Pulumi), On-premises (VPS, NAS), Hybrid cloud, FinOps (cloud cost optimization)
Observability: Logs, metrics & tracing (OpenTelemetry)
Delivery and quality: CI/CD (GitHub Actions), Testing (pytest)
Languages: Python, SQL, Go, TypeScript, C++
Architecture and design: DDD, Hexagonal architecture, SOLID, BDD
Security and identity: Identity & access management (Keycloak, OIDC / OAuth2, RBAC)
Industrial IoT: PLC programming (Structured Text, IEC 61131-3, matiec)
Leadership: Technical strategy, Engineering team management, Hiring, Agile methodology (Scrum, roadmaps), Engineering standards
Compliance: ISO/IEC 42001 (AI management), ISO/IEC 27001 (information security), ENS (Spanish National Security Framework), GDPR (data protection), EU AI Act (European AI regulation)

Sectors

Wildfire fighting (applied AI): Real-time asset monitoring: aircraft, crews and vehicles (asset tracking); Video surveillance with live aerial video streaming, in suppression and in prevention; Mission management and resource allocation; Synchronization over intermittent connectivity (offline-first); Computer vision on aerial imagery; Burned area estimation (burned area mapping); Aircraft and vehicle telemetry (IoT, MQTT).

Forest management (applied AI): Forestry contract management; Activity and machinery reports (field operations); Asset monitoring: machinery position, status and history (fleet tracking); Machinery video surveillance and telemetry (IoT, live video); Plant and weed detection in aerial imagery (computer vision, object detection); Georeferencing and mosaics (geospatial, GIS); Multispectral imagery (remote sensing); Hybrid on-premises / cloud architecture.

Emergency management (applied AI): Mission management and resource allocation; Automatic plan generation (automated planning); Incident tracking (incident management); Video surveillance and telemetry of field resources (live video, IoT).

Fertigation: Precision agriculture; Irrigation and nutrient control (industrial automation); Crop and soil sensor data and telemetry (IoT, time series).